

Contribution Statement

ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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Both team members contributed meaningfully to this project. Responsibilities and completed tasks are listed below.

Yahor Lahunovich

Architected the core ChartPRM pipeline on **CharXiv**, engineered custom VLM alignment trainers, and authored the foundational report.

- Curated the 500-question CharXiv reasoning pool (8 domains, 62 chart types) and established the protected 100-question holdout.
- Designed the structured reasoning protocol for Qwen2.5-VL-3B and evaluated 1,274 rollouts via muse-spark-1.1 with cost-saving rollout batching.
- Conducted semantic error analysis across 2,920 refuted steps, establishing the 9-category failure taxonomy that uncovered the perceptual bottleneck and grounded the reward tree.
- Engineered minimal, native PyTorch training engines for SFT, Full DPO, Step-DPO, KTO, and SFT→DPO, bypassing TRL VLM memory bugs to train within a single 16GB T4 GPU envelope.
- Benchmarked all alignment variants on the holdout, diagnosing and resolving policy log-prob collapse and formatting trade-offs.
- Built the centralized scientific visualization framework (SciencePlots, Paul Tol palettes) generating all 29 benchmark and diagnostic figures.
- Set up the ACL LaTeX layout, authored the Introduction, Related Work, ChartPRM Methods, and Results & Analysis sections, and curated the bibliography.
- **AI usage summary:** Used **Antigravity** and **Cursor** as AI assistants for writing and validating all codebase implementation, and for drafting report sections (with manual modifications). All research ideas, hypotheses, experimental formulations, and empirical analyses were conceived and conducted entirely by me.

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Presented Yin et al.'s *Dynamic and Generalizable Process Reward Modeling* (DG-PRM) for the seminar and directed its adaptation, plus a cross-dataset study.

- Directed the reward-tree construction: 33 scored criteria under 9 categories, from clustered judge critiques.
- Directed the dynamic multi-criteria judge (scored via gemini-3.5-flash-lite, distinct from the muse-spark-1.1 binary judge) and Pareto-dominance-filtered pair selection for DPO (**Pareto-DPO**).
- Directed the reference-free **SimPO** baseline implementation.
- Directed the ChartGemma specialist-baseline and oracle-ceiling diagnostic experiments.
- Designed and ran the cross-dataset generalization test on **ChartQA**: sampling, rollout generation, unmodified reward-tree transfer, and retraining/evaluating all seven methods on a 30-question holdout.
- Authored the DynamicPRM, Test-Time Verification and Search, and Cross-Dataset Generalization (ChartQA) sections, and their tables/figures; also revised the Abstract, Introduction, and ChartPRM Framework as the project's scope grew to two benchmarks and seven methods.
- **AI usage summary:** Used **Claude Code** (Anthropic) as an AI coding agent for essentially all of the code and analysis above (reward tree, dynamic judge, SimPO, ChartQA pipeline) and for drafting my report sections. I directed every experiment (what to try, when to pivot), set the evidentiary bar (e.g., requiring bootstrap significance tests before claiming a win), and personally verified and interpreted all results before they went into the report.

Both authors confirm this statement accurately reflects their respective contributions to the ChartPRM project.